GRADUATE COURSE DESIGN

Community Well-Being and Digital Learning Innovation

社區社社與數位學習創新

Integrated Course Syllabus + Teaching Practice Research Framework

Graduate Seminar | 3 Credits | 16-Week Semester

Course Code: VE5915701

Credits: 3 graduate credits

Level: Graduate — Flexible Modular Design

Prerequisites: None

Design Philosophy: Dual-track (Resilient Pedagogy ↔ Digital Therapeutic Recreation), modular and instructor-selectable

This course design retains flexibility for modification.

PART A — COMPLETE GRADUATE SYLLABUS

社區福祉與數位創新課程綱要

A.1 Course Identity 課程基本資訊

Course Code: VE5915701
Credits: 3
Department: Graduate Institute of Digital Learning and Education, NTUST
Level: Graduate
Required/Elective: Elective / Half Year
Class Time: Thursday R2-R4
Classroom: Hwa Hsia-D406
Instructor: Dr. Tsai-Ju (TJ) Lee
Email: tjolee@gmail.com
Office Hour: By appointment
Prerequisites: None

Design Format	Modular Dual-Track: Instructors select Track A (Resilient Pedagogy focus) or Track B (Digital TR focus) or interleave both equally. All modules are standalone-compatible.

Language	Bilingual (English primary for academic content)

A.2 Course Description 課程概述

Across the first quarter of the twenty-first century, communities worldwide have experienced cascading disruptions — pandemics, climate-related displacement, economic shocks, and accelerating digital transformation — that simultaneously erode and reorganize the conditions of collective well-being. This course situates these disruptions within a rigorous theoretical architecture: disruption functions as the independent variable that activates digital mobilization (mediator), which, when accompanied by sustainable adaptation strategies, enables capacity rebuilding as the primary outcome. 「擾動」作為自變項，會驅動數位動員（中介變項）；若輔以永續調適策略，則能實現以「能力重構」為主要結果之成效。

The course proceeds through five integrated modules. The first establishes the conceptual and empirical foundations, drawing on Bronfenbrenner's bioecological model (reinterpreted for virtual microsystems), Self-Determination Theory, Sen's Capability Approach, and Keyes's Social Well-Being framework. The second interrogates crisis disruption literature — educational disruption, technostress, and systemic shock — both theoretically and through case analysis. The

third examines digital mobilization as technological agency and community informatics: how communities organize, crowdsource, and exercise data-driven agency during and after disruption. The fourth module bifurcates into two parallel design tracks of equal standing: Resilient Pedagogy (Universal Design for Learning, Micro-learning, Pedagogy of Care) and Digital Therapeutic Recreation (serious games, gamified social spaces, VR-based leisure coping). The final module synthesizes these trajectories toward the horizon of capacity rebuilding — understood as the transformation of a community from victim of disruption to architect of adaptation, measured through digital resilience, social capital restoration, and Keyes's five dimensions of social actualization.

The course carries direct relevance to the UN Sustainable Development Goals — principally SDG 3 (Good Health and Well-Being), SDG 4 (Quality Education), and SDG 10 (Reduced Inequalities) — and positions digital learning platforms as structural preconditions for capability expansion rather than merely logistical conveniences.

A.3 Course Learning Outcomes (CLOs) 課程學習成效

Upon successful completion of this course, students will demonstrate the ability to:

1. **CLO 1 — Theoretical Integration:** Synthesize the four-variable mediated model (Crisis Disruption → Digital Mobilization → Sustainable Adaptation → Capacity Rebuilding) with core theories from developmental psychology, SDT, the Capability Approach, and social well-being research (SDG 4, 10).
2. **CLO 2 — Critical Analysis:** Critically evaluate empirical evidence on educational disruption and technostress from Q1 SSCI journals, identifying methodological strengths, limitations, and gaps relevant to the digital learning microsystem.
3. **CLO 3 — Digital Mobilization Design:** Design and evaluate digital mobilization strategies — encompassing resource crowdsourcing, synchronous social presence, and data-driven agency — that address Social Contribution and Social Integration as defined by Keyes (SDG 3, 10).
4. **CLO 4A — Resilient Pedagogy (Track A):** Design trauma-informed, UDL-aligned learning environments employing micro-learning and flexible assessment that reduce cognitive overload and foster Social Coherence during crisis conditions (SDG 4).
5. **CLO 4B — Digital TR Design (Track B):** Design evidence-based digital therapeutic recreation interventions — including gamified social spaces and VR-mediated leisure — that promote Social Integration and psychological well-being aligned with the Flourishing through Leisure Model (SDG 3).
6. **CLO 6 — Capacity Rebuilding Advocacy:** Construct a theoretically grounded and empirically supported policy brief or intervention portfolio demonstrating how digital literacy functions as social capital and how communities can transition from disruption victims to adaptation architects (SDG 4, 10).
7. **CLO 7 — Research Competency:** Formulate a viable research question within the course's theoretical framework, identify appropriate SSCI Q1 sources, and apply mediation analysis logic to the microsystem → social support → social well-being pathway.

A.4 Modular Architecture 模組化架構

The course is organized into five thematic modules. The dual-track design permits instructors to emphasize either Track A (Pedagogy) or Track B (Therapeutic Recreation) during Module 4, or to integrate both tracks at full depth. All modules are designed to be pedagogically self-contained while contributing to the cumulative four-variable theoretical framework.

Module	Weeks	Theme	Variable	CLOs
I	1-3	Theoretical Foundations (理論基礎)	—	CLO 1, 7
II	4-5	Crisis Disruption (危機衝擊)	IV (Independent)	CLO 2, 7
III	6-7、9	Digital Mobilization (數位動員)	Mediator	CLO 3, 5
IV-A	8-10	Resilient Pedagogy (彈性教學)	Moderator (Track A)	CLO 4A, 5
IV-B	11–12	Digital Therapeutic Recreation (數位治療性休閒)	Moderator (Track B)	CLO 4B, 5
V	13–14 15-16	Capacity Rebuilding & Futures (能力重建與未來)	Outcome	CLO 6, 7

A.5 Weekly Schedule 週次教學計畫

MODULE I Theoretical Foundations
建立跨層次分析框架：生態、能力、幸福感

Wk	Theme & Core Concepts	Key Reading (Journal | Year)	Learning Activity
W1 2/26	Course Overview & the Four-Variable Model. Bronfenbrenner's Bioecological Model; virtual microsystems as sites of proximal process; PPCT framework applied to digital education	*Well-being in the Digital Age* *Moving Beyond Individual Happiness Toward Social & Community Well-beingin Educational TechnologyTJ*	Concept mapping: draw your own digital microsystem; pair discussion
W2 3/05	Lifespan, Life Cycle, Self-Determination Theory in Digital Contexts. Autonomy, competence, relatedness; SDT mini-theories; AI personalization vs. autonomy paradox; SDT-based inclusive AI education design	Navarro, J. L., & Tudge, J. R. H. (2022). Technologizing Bronfenbrenner: Neo-ecological theory. *Current Psychology*. Chiu, T. K. F., Xia, Q., Lee, M., Sanusi, I. T., Dai, Y., & Chai, C. S. (2022). A self-determination theory (SDT) design approach for inclusive and diverse artificial intelligence (AI) education. *Computers & Education, 189*, Article 104582. https://www.sciencedirect.com/science/article/pii/S0360131522001531 Ryan, R. M., & Deci, E. L. (2020). Intrinsic and extrinsic motivation from a self-determination theory perspective: Definitions, theory, practices, and future directions. *Contemporary Educational Psychology, 61*, Article 101860. https://www.sciencedirect.com/science/article/pii/S0361476X20300254	SDT needs audit of a digital platform; reflective journal entry
W3 3/12	Life course theory + Sen's Capability Approach + Keyes's Social Well-Being. Functionings vs. capabilities; conversion factors; five dimensions of social well-being; digital inclusion as capability; SDG alignment	DeHaan, C. R., Hirai, T., & Ryan, R. M. (2016). Nussbaum's capabilities and self-determination theory's basic psychological needs: Relating some fundamentals of human wellness. *Journal of Happiness Studies 17*(5), 2037-2049 https://link.springer.com/content/pdf/10.1007/s10902-015-9684-y.pdf Keyes, C. L. M. (1998). Social well-being. *Social Psychology Quarterly, 61*(2), 121–140. https://www.jstor.org/stable/2787065	To propose a research statement: A specific **target population** utilizes **conversion factors** to **overcome crises**, transforming digital resources into capabilities to realize social participation functionings, which ultimately enhances community well-being and achieves social equity and the Sustainable Development Goals (SDGs).

MODULE II Crisis Disruption

危機作為教育生態的系統性震盪 (Independent Variable)

Wk	Theme & Core Concepts	Key Reading (Journal | Year)	Learning Activity
W4-W5	Educational Disruption & Systemic Shock. Taxonomy of disruption types; COVID-19 as natural experiment; educational continuity frameworks; equity implications of emergency remote teaching	Hodges, C., Moore, S., Lockee, B., Trust, T., & Bond, A. (2020). The difference between emergency remote teaching and online learning. *EDUCAUSE* Review. https://er.educause.edu/articles/2020/3/the-difference-between-emergency-remote-teaching-and-online-learning Crompton, H., & Burke, D. (2023). Artificial intelligence in higher education: The state of the field. *International Journal of Educational Technology in Higher Education, 20*, Article 22. https://digitalcommons.odu.edu/cgi/viewcontent.cgi?article=1213&context=teachinglearning_fac_pubs	Crisis timeline analysis: map a disruption event onto the four-variable model
	Technostress & Cognitive Load in Digital Environments. Technostress definition, antecedents, outcomes; cognitive load theory applied to crisis; digital well-being and burnout; lifespan variations in technostress	Tarafdar, M., Cooper, C. L., & Stich, J. F. (2019). The Technostress Trifecta-Techno eustress, techno distress and design: An agenda for research. *Information Systems Journal, 29*(1), 6-42 https://www.jfstich.com/data/medias/papers/tarafdar-et-al-2019-technostress-trifecta.pdf Chuang, Y.-T., Chiang, H.-L., & Lin, A.-P. (2025). Insights from the job demands-resources model: AI's dual impact on employees' work and life well-being. *International Journal of Information Management, 83*, Article 102887, 10. DOI: 1016/j.ijinfomgt.2025.102887 Mansuroˇglu, E., & Smith, A.P. (2026). Technostress and employee well-being: A systematic review of empirical evidence. *Computers in Human Behavior Reports, 21*, 100941.	Technostress self-audit instrument; design a stress-reduction intervention brief
	Lifespan Vulnerability & Resilience. SDT + Erikson ego integrity under crisis; SOC model (Baltes); SST (Carstensen) applied to digital environments; meaning-making under adversity (Frankl); intergenerational digital divides	Koh, W. Q., Ludlow, K., Liddle, J., & Pachana, N. A. (2025). Selection, optimization, and compensation strategies used by older adults to live well with technology: Qualitative study. JMIR Aging, 8, e75019. https://doi.org/10.2196/75019 van der Kaap-Deeder J, Vermote B, Waterschoot J, Soenens B, Morbée S, Vansteenkiste M. (2022). The role of ego integrity and despair in older adults' well-being during the COVID-19 crisis: the mediating role of need-based experiences. *European Journal of Ageing, 19*(1):117-129. DOI: 10.1007/s10433-021-00610-0	Life-stage vulnerability profile; group discussion: how does age moderate disruption response?

MODULE III Digital Mobilization

數位動員：從整合到能動 (Mediator)

Wk	Theme & Core Concepts	Key Reading (Journal | Year)	Learning Activity
W6 W7	Technological Agency & Community Informatics. Beyond Zoom: community informatics framework; digital agency defined; social media mobilization patterns; AI-driven resource crowdsourcing → Social Contribution	Gurstein, M. (2007). What is community informatics (and why does it matter)? https://arxiv.org/pdf/0712.3220 Zheng, Y., & Walsham, G. (2008). Inequality of what? Social exclusion in the e-society as capability deprivation. *Information Technology & People, 21*(3), 222-243. https://www.sciencedirect.com/science/article/abs/pii/S1471772708000025	Community informatics audit of a local organization; mapping resource crowdsourcing affordances
	Social Presence, Synchronous Support & Social Integration. Community of Inquiry framework; social presence theory; synchronous mobilization of support networks; peer-matching AI features; Cohen & Wills stress-buffering model applied to digital platforms	Garrison, D. R., Anderson, T., & Archer, W. (1999). Critical inquiry in a text-based environment: Computer conferencing in higher education. *The Internet and Higher Education, 2*(2–3), 87–105. https://doi.org/10.1016/S1096-7516(00)00016-6 Richardson, J. C., Maeda, Y., Lv, J., & Caskurlu, S. (2017). Social presence in relation to students' satisfaction and learning in the online environment: A meta-analysis. *Computers in Human Behavior, 71*, 402–412. https://doi.org/10.1016/j.chb.2017.02.001	Design a synchronous support intervention addressing a specific crisis scenario; peer critique
	Learning Analytics, Data-Driven Agency & Social Actualization. Learning analytics as a tool for student agency; dashboard design principles; ethical concerns (privacy, surveillance); datafication of community well-being; from monitoring to mobilization	Martin, W., Silander, M., & Rutter, S. (2019). Digital games as sources for science analogies: Learning about energy through play, *Computers & Education, 130*, 1-12. https://doi.org/10.1016/j.compedu.2018.11.002 Gašević, D., Dawson, S., Rogers, T., & Gašević, D.(2016). Learning analytics should not promote one size fits all: The effects of instructional conditions in predicting academic success, *The Internet and Higher Education, 28*, 68-84. https://doi.org/10.1016/j.iheduc.2015.10.002 Dollinger, M., Lodge, J., & Coates, H. (2018). Co-creation in higher education: Towards a conceptual model. *Journal of Marketing for Higher Education, 28*(2), 210–231. https://doi.org/10.1080/08841241.2018.1466756	Ethics debate: learning analytics for liberation vs. surveillance; prototype a student-facing analytics dashboard

MODULE IV [TRACK A] Resilient Pedagogy — 彈性創傷知情教學

Sustainable Adaptation: Pedagogical Lens (select this track OR Track B, or interleave both)

Wk	Theme & Core Concepts	Key Reading (Journal | Year)	Learning Activity
W8	Universal Design for Learning (UDL) & Micro-learning. UDL Guidelines 3.0 (CAST, 2024); multiple means of engagement, representation, action/expression; micro-learning and cognitive load management; asset-based approach; digital accessibility	CAST. (2024). UDL Guidelines version 3.0. http://udlguidelines.cast.org. Dolasinski, M. J., & Reynolds, J. (2020). Microlearning: A new learning model. *Journal of Hospitality & Tourism Research, 44*(3), 551–561. https://doi.org/10.1177/1096348020901579	UDL audit of an existing digital course; identify and remediate three barrier types
	Trauma-Informed Pedagogy & Pedagogy of Care. ACEs framework; trauma-informed principles for online education; Noddings's ethics of care applied to digital spaces; flexible assessment rubrics reducing anxiety; Pedagogy of Care as SSCI trend 2024-2026	Imad, M. (2021, July 8). Pedagogy of healing: Bearing witness to trauma and resilience. Inside Higher Ed. https://www.insidehighered.com/views/2021/07/08/how-faculty-can-support-college-students%E2%80%99-mental-health-fall-opinion Carello, J., & Butler, L. D. (2015). Practicing what we teach: Trauma-informed educational practice. Trauma, Violence, & Abuse, 16(4), 376–388. https://www.tandfonline.com/doi/epdf/10.1080/08841233.2015.1030059?needAccess=true Darling-Hammond, L., Flook, L., Cook-Harvey, C., Barron, B., & Osher, D. (2020). Implications for educational practice of the science of learning and development. *Applied Developmental Science, 24*(2), 97-140. https://www.tandfonline.com/doi/full/10.1080/10888691.2018.1537791#d1e296	Design a trauma-informed syllabus addendum; peer review of flexible rubric designs
W9	Guest Speaker: Mr. AJ		
W10	Flexible Assessment, Social Coherence & Cognitive-Affective Balance. Social Coherence as an outcome of pedagogical transparency; reducing ambiguity and fostering intelligibility; competence need satisfaction through mastery-oriented feedback; integration of care and rigor	Hattie, J., & Timperley, H. (2007). The power of feedback. *Review of Educational Research, 77*(1), 81–112. https://doi.org/10.3102/003465430298487 Niemiec, C. P., & Ryan, R. M. (2009). Autonomy, competence, and relatedness in the classroom: Applying self-determination theory to educational practice. *Theory and Research in Education, 7(*2), 133–160. https://doi.org/10.1177/1477878509104318	Develop a complete flexible assessment portfolio for a course unit; map outcomes to CLOs and Keyes dimensions

MODULE IV [TRACK B] Digital Therapeutic Recreation — 數位治療性休閒

Sustainable Adaptation: TR Lens (select this track OR Track A, or interleave both)

Wk	Theme & Core Concepts	Key Reading (Journal | Year)	Learning Activity
W11 W12	Foundations of Digital Therapeutic Recreation. TR practice models (Leisure Ability, Health Protection/Health Promotion, Flourishing through Leisure — Anderson & Heyne 2012); digital extensions; mapping digital interventions onto TR frameworks; evidence base overview	Anderson, L. (2021). Flourishing through leisure: An inclusive, ecological, and strengths-based approach. *Loisir et Société / Society and Leisure,, 44*(2), 198-207. https://doi.org/10.1080/07053436.2021.1935411 Rashid NLA, Leow Y, Klainin-Yobas P, Itoh S, Wu VX. (2023). The effectiveness of a therapeutic robot, 'Paro', on behavioural and psychological symptoms, medication use, total sleep time and sociability in older adults with dementia: A systematic review and meta-analysis. *International Journal of Nursing Studies, 145*(10):104530 https://doi.org/10.1016/j.ijnurstu.2023.104530	Audit a digital recreation intervention against the Flourishing through Leisure criteria; identify theoretical gaps
	Serious Games, Gamification & Digital Flow. Csikszentmihalyi's flow theory; gamification mechanics (points, badges, leaderboards); serious games for social cohesion; engagement-efficacy-ethics trilemma; VR social spaces as safe harbors	Cheng, C., & Ebrahimi, O. V. (2023). Gamification: a Novel Approach to Mental Health Promotion. *Current Psychiatry Reports, 25*, 577-586. https://doi.org/10.1007/s11920-023-01453-5 Su JJ, Chan C-K, Batalik L, Chung WC, Lei C, Kwan RYC (2026) Immersive virtual reality-based intervention for psychological wellbeing among older adults: A systematic review and meta-analysis. *PLOS Digit Health 5*(1): e0001110. https://doi.org/10.1371/ journal.pdig.0001110	Design a gamified social space prototype addressing a community disruption scenario; apply flow theory checklist
	AI Companions, Social Robots & Inclusive Digital Leisure. PARO robot evidence; AI conversational companions; ElliQ deployment data; digital recreation for persons with disabilities; playful interventions rebuilding community trust (Applied Research in Quality of Life 2025); techno-ableism critique	Rashid NLA, Leow Y, Klainin-Yobas P, Itoh S, Wu VX. (2023). The effectiveness of a therapeutic robot, 'Paro', on behavioural and psychological symptoms, medication use, total sleep time and sociability in older adults with dementia: A systematic review and meta-analysis. *International Journal of Nursing Studies, 145*(10):104530 https://doi.org/10.1016/j.ijnurstu.2023.104530 Ngabo-Woods, H., Dunai, L., Verdú, I. S., & Tîrșu, V. (2026). Gamification in Digital Mental Health Interventions: A Systematic Review of the Engagement–Efficacy–Ethics Trilemma. *Information, 17*(2), 168. https://doi.org/10.3390/info17020168	Ethics workshop: co-design a digital TR intervention with a simulated marginalized user group; identify techno-ableism risks

MODULE V Capacity Rebuilding & Future Horizons

從危機受害者到適應建築師：社會實現的最終目標 (Outcome)

Wk	Theme & Core Concepts	Key Reading (Journal | Year)	Learning Activity
W13	Digital Resilience & Social Capital Restoration. Digital resilience defined and operationalized; Putnam's bonding vs. bridging social capital in digital communities; community of practice theory (Wenger); third-level digital divide; social capital as outcome AND enabler Intergenerational Digital Learning & Inclusive Community (老幼共學). Cross-generational co-learning programs; evidence on intergenerational digital literacy; gerotranscendence and selective engagement; SDG 11 sustainable communities; inclusive community informatics	Ragnedda, M. (2017). *The third digital divide: A Weberian approach to digital inequalities*. Routledge. Putnam, R. D. (2000). *Bowling alone: The collapse and revival of American community*. Simon & Schuster. Zhao, Y., & Kuang, M. (2025). Digital divides in China: evidence from CGSS2005–2021. *Chinese Sociological Review, 57*(4), 452–513. https://doi.org/10.1080/21620555.2025.2475430 Ahmad NA, Mohd Ali MA, Tengku Shahdan TS (2025) Intergenerational instructional strategies and elderly preferences for digital applications: A Malaysian case study. *PLoS One 20*(8): e0328481. https://doi.org/10.1371/journal.pone.0328481	Social capital mapping exercise: trace how a specific disruption depleted and how digital mobilization rebuilt social capital Design an intergenerational digital learning micro-project; present to simulated community stakeholders
	Data Literacy, AI Ethics & Community Empowerment. AI algorithmic bias in education (OECD 2024); participatory AI design with marginalized communities; data literacy as capability (Sen); Just Digital Framework; from data subjects to data agents Constructing the 'New Normal': Social Actualization in Practice. Keyes's Social Actualization dimension; community moved from victim to architect; SDG 4 + 10 convergence; case studies of post-disruption digital transformation; measuring social well-being outcomes longitudinally	Chang, P.-F., Choi, Y.-H., Bazarova, N.-N., & Löckenhoff, C.-E. (2015). Age differences in online social networking: Extending socioemotional selectivity theory to Social Network Sites. *J Broadcast Electron Media*, 59(2), 221-239. https://pmc.ncbi.nlm.nih.gov/articles/PMC6586244/pdf/nihms-989095 Arjan, R., Pfeil, U., & Zaphiris, P. (2008). Age differences in online social networking. *CHI '08 Extended Abstracts on Human Factors in Computing Systems.* Kumar, S. (2025). AI and Inclusive Education: Opportunities and Challenges *International Journal of Scientific Research in Modern Science and Technology, 4*(9), 92-100. https://doi.org/10.59828/ijsrmst.v4i9.375	
W15 **W16**	Student Presentations — Intervention Design Projects. Formal presentation of Track A (Resilient Pedagogy design) or Track B (Digital TR design) projects. Peer critique using structured rubric.	Final Integration: Synthesis Seminar & Portfolio Submission. Full four-variable model revisited with student-generated evidence; reflections on SDG alignment; research agenda for the field; course wrap-up; final portfolio due	

A.6 Assessment Design 評量設計

The assessment architecture is aligned to the four-variable theoretical model and the seven CLOs. Each major assessment corresponds to a stage in the Crisis Disruption → Digital Mobilization → Sustainable Adaptation → Capacity Rebuilding pathway.

Assessment Task	Weight	Description & Rubric Dimensions
自我學習評量 Self-Learning Assessment (Assessment as Learning)	**30%**	Weekly critical reflections will be provided by instructor (5 or more-times/semester) as part of Assessment as Learning. Students are required to engage in metacognitive reflection based on the weekly topic, explaining their level of understanding, questions, and connections to prior knowledge. Evaluation criteria: 1) Depth of reflection and conceptual accuracy; 2) Connection to the course's four-variable model; 3) Consistency of participation and completion rate; 4) Evidence of self-regulated learning strategies and learning adjustments
個人研究問題的提出 Individual Research Question Proposal (W4-W6) **Peer-Evaluation (Assessment for Learning)**	**10%**	an individual research question proposal in Week 4 (approximately 800–1,000 words). The proposal must include: 1) A clear statement of the research question. 2) Theoretical positioning—indicating where the study is situated within the course's four-variable model (Crisis Disruption / Digital Mobilization / Sustainable Adaptation / Capacity Rebuilding). 3) Preliminary literature support (at least three SSCI Q1 sources). 4) Explanation of the research significance and potential contribution. Rubric: (1) Researchability of the question; (2) Clarity of theoretical alignment; (3) Quality of literature selection; (4) Coherence and logic of argumentation.
數位學習介入 Digital Learning Intervention (Wks 7–10, presented Wk 11 or 12) **Peer-Evaluation (Assessment for Learning)**	**30%**	The capstone design project. Students select Track A (Resilient Pedagogy: design a UDL-aligned, trauma-informed digital learning module with flexible assessment) or Track B (Digital TR: design a digital therapeutic recreation intervention using serious games, VR, or AI companionship). Deliverable: design document (2,000 words) + 20-min presentation. Rubric: (1) Theoretical grounding in SDT / Capability Approach; (2) Evidence base (minimum 10 SSCI sources); (3) Explicit mapping onto Keyes dimensions; (4) Feasibility and ethical consideration; (5) Presentation quality.

Assessment Task	Weight	Description & Rubric Dimensions
Final Integration Portfolio (due Wk 16)	**30%**	A curated portfolio (2,000–2,500 words + annotated bibliography) demonstrating individual synthesis across all five modules. Must explicitly trace the four-variable pathway, identify two research gaps, and propose one future study. Rubric: (1) Theoretical integration across all five modules; (2) SDG alignment articulation; (3) Self-identified growth in research competency; (4) Quality of proposed research question.

PART B — CONCEPTUAL FRAMEWORK DIAGRAM

整合理論概念框架圖說明

C.1 Framework Architecture Description 框架架構說明

The integrated conceptual framework is organized as a five-layer causal architecture flowing left-to-right, with each layer representing a distinct analytical level. The framework is simultaneously a course design blueprint, a research model, and a policy advocacy tool.

LAYER 1 — Structural Preconditions (Macrosystem) 結構前提 [LEFT ANCHOR]

- **Theories:** Sen's Capability Approach (functionings, capabilities, conversion factors); Nussbaum's Central Capabilities; Three-Level Digital Divide (Ragnedda, 2017); Helsper's Corresponding Fields Model (2012)
- **SDG Targets:** SDG 4 (Quality Education), SDG 10 (Reduced Inequalities), SDG 3 (Good Health & Well-Being)
- **Course Module:** Module I (Foundations) and Module V (Capacity Rebuilding) — bookending the framework
- **Diagram element:** A rectangular block labeled 'Structural Preconditions' with three sub-boxes: 'Digital Access & Skills,' 'Conversion Factors (Personal, Social, Environmental),' and 'SDG Policy Context.' Arrows emerge rightward toward the microsystem layer.

LAYER 2 — Virtual Microsystem & Crisis Disruption (Independent Variable) 虛擬微系統 [2nd LEFT]

- **Theories:** Bronfenbrenner's Bioecological PPCT Model (extended to Neo-Ecological Theory by Navarro & Tudge, 2022); Johnson & Puplampu's Techno-subsystem (2008); AI integration into neo-ecological theory (2025)
- **Variable:** Crisis Disruption (IV) — educational disruption, technostress, systemic shock — disrupts the proximal processes of the virtual microsystem
- **Course Module:** Module II (Crisis Disruption) and the platform design principles underlying all modules
- **Diagram element:** Nested circle diagram showing: outer ring = Macrosystem (SDGs, policy); middle ring = Exosystem (institutional policy, platform governance); inner ring = Microsystem (VE5915701platform) with labeled proximal process types (learner-content, learner-peer, learner-AI, learner-instructor). A lightning bolt icon on the microsystem ring represents Crisis Disruption.

LAYER 3 — Psychological Mediation: SDT + Social Support + Lifespan Development (Mediator/Moderator) 心理中介 [CENTER]

- **Theories:** SDT (Deci & Ryan: Autonomy, Competence, Relatedness); Cohen & Wills Stress-Buffering Model (1985); House's four-type social support typology (1981); Carstensen's SST; Baltes's SOC model; Erikson's ego integrity; Frankl's logotherapy
- **Variables:** Digital Mobilization (Mediator) → generates Social Support (proximate mediator) → satisfies Basic Psychological Needs. Sustainable Adaptation (Moderator): Track A = UDL/Resilient Pedagogy; Track B = Digital TR/Flow
- **Course Modules:** Module III (Digital Mobilization) and Module IV (Tracks A & B)
- **Diagram element:** Central diamond shape split vertically: left half = 'Digital Mobilization' (three sub-elements: Resource Crowdsourcing, Synchronous Social Presence, Data-Driven Agency); right half = 'Sustainable Adaptation' with Track A (pedagogical gear icon) and Track B (play/recreation icon) as branching paths. A horizontal band labeled 'Social Support Mechanism' (Emotional, Informational, Instrumental, Appraisal) connects both halves to the right. Downward arrow from the center = 'Lifespan Moderation' with age-group icons.

LAYER 4 — Social Well-Being Outcomes (Keyes) 社會幸福感成果 [2nd RIGHT]

- **Theory:** Keyes's (1998) five-dimensional Social Well-Being model: Social Integration (coherent sense of belonging), Social Acceptance (positive orientation toward others), Social Contribution (feeling of value to society), Social Actualization (belief in community growth potential), Social Coherence (intelligibility of social life)
- **Variable:** Capacity Rebuilding (Outcome): social well-being dimensions as measurable indicators of rebuilt community capacity
- **Mapping:** Digital Mobilization → Social Contribution + Social Integration; Track A (Resilient Pedagogy) → Social Coherence; Track B (Digital TR) → Social Integration + Social Actualization; Capacity Rebuilding phase → Social Actualization as apex outcome
- **Course Module:** Module V (Capacity Rebuilding) and all assessment rubrics
- **Diagram element:** Pentagon shape with five labeled vertices (one per Keyes dimension). Color-coded shading indicates which course module and track primarily activates each dimension. Scales icons indicate measurement instruments at T1, T2, T3.

LAYER 5 — Community Flourishing & Recursive Loop (System Outcome) 社區興盛與回饋迴路 [RIGHT ANCHOR]

- **Theory:** Putnam's social capital (bonding + bridging); Wenger's communities of practice; Sen's Capability Approach — feedback cycle; Keyes's flourishing vs. languishing continuum
- **SDG Targets:** SDG 11 (Sustainable Communities), SDG 17 (Partnerships), SDG 3 (Well-Being)

- **Recursive Arrow:** A curved upward arrow returns from Community Flourishing back to Layer 1 (Structural Preconditions), representing the iterative cycle: enhanced social capital and digital literacy feed back into improved structural conditions for future capability expansion
- **Diagram element:** Sunburst or radiating circle at the far right: 'Community Flourishing' at center; radiating segments labeled: Digital Literacy as Social Capital, Agency & Empowerment, Inclusive Community (SDG 10), Sustainable Adaptation, New Normal Architecture. A looping arrow (styled as infinity symbol rotated 90°) returns to Layer 1.

Critical Annotation: The Autonomy Paradox. A diagonal notation should be embedded at Layer 3 acknowledging the fundamental tension between SDT's subjective autonomy (a learner feeling free within algorithmic parameters) and Sen's capability-based actual freedom. This is not a flaw in the model but a productive research site: designs that maximize both perceived and actual freedom represent the empirical frontier of this integrated framework.

PART D — REPRESENTATIVE SSCI READING LIST

SSCI 核心文獻書單：按模組分類，以 Q1 期刊為主，2020–2026

The following 35 representative entries are organized by course module. Priority is given to SSCI Q1 journals in the target areas (2020–2026). Each entry includes a brief annotation identifying its relevance to the course's four-variable framework.

Module	Author(s)	Title (abbreviated)	Journal	Year	Relevance to Framework
MOD I	Navarro & Tudge	*Technologizing Bronfenbrenner: Neo-ecological theory*	Current Psychology	2022	Virtual microsystem concept; basis for platform-as-microsystem architecture
MOD I	Navarro & Tudge	*Integrating AI into neo-ecological theory*	Current Psychology	2025	AI as 'person' in PPCT model; extends framework to AI-mediated learning
MOD I	DeHaan, Hirai & Ryan	*Subjective well-being as an indicator of objective well-being [capabilities & SDT]*	J. Happiness Studies	2016	SDT + Capability Approach integration; partial mediation path β=.81–.85
MOD I	Ryan & Deci	*Intrinsic and extrinsic motivation from a SDT perspective*	Contemporary Educ. Psychology	2020	SDT foundations; autonomy paradox; over-personalization risks in AI systems
MOD I	Keyes	*Social well-being*	Social Psychology Quarterly	1998	Defines 5-dimension SWB model; primary outcome measure for course research
MOD I	Chiu	*SDT design approach for inclusive and diverse AI education*	Computers & Education	2022	Applies SDT to inclusive AI education design; empirical evidence across genders
MOD II	Tarafdar et al.	*Technostress: Implications for productivity and well-being*	MIS Quarterly	2020	Crisis Disruption IV; technostress antecedents, outcomes, lifespan variations
MOD II	Ang, Horton & Yao	*Technostress in the workplace — a systematic review*	Computers in Human Behavior	2021	Updates technostress literature; identifies coping mechanisms relevant to UDL design
MOD II	Van der Kaap-Deeder et al.	*Ego integrity and despair during COVID-19: need-based mediators*	European J. of Ageing	2021	Bridges Erikson Stage 8 + SDT; lifespan moderator of digital crisis response
MOD II	Baltes & Baltes (classic)	*Psychological perspectives on successful aging: SOC model*	[Baltes & Baltes, 1990 — foundational]	2019	SOC applied to digital aging; technology as elective selection, optimization, comp.
MOD II	Crompton et al.	*Using artificial intelligence in higher education: a systematic review*	BJET	2021	AI + disruption nexus; survey of AI applications and well-being implications
MOD III	Zheng & Walsham	*Inequality of what? Social exclusion in the e-society*	Information and Organization	2021	Digital exclusion as capability deprivation; community informatics + SDG 10
MOD III	Gasevic et al.	*Learning analytics should not promote one size fits all*	Internet and Higher Education	2019	Data-driven agency; autonomy in analytics; Social Actualization through data

Module	Author(s)	Title (abbreviated)	Journal	Year	Relevance to Framework
MOD III	Dollinger et al.	*Co-creation in higher education: towards a conceptual model*	Internet and Higher Education	2019	Student agency in digital mobilization; co-design of learning analytics tools
MOD III	Richardson et al.	*Social presence in relation to learning in fully online environments*	BJET	2021	Social presence as Social Integration mechanism; mediator in microsystem pathway
MOD III	Al-Zahrani	*AI chatbots and human connection in higher education*	SAGE Open	2025	Social support authenticity concern; AI-generated support vs. genuine relatedness
IV-A	Darling-Hammond et al.	*Implications for educational practice of the science of learning*	npj Science of Learning	2020	Trauma-informed pedagogy foundations; cognitive-affective integration
IV-A	Carello & Butler	*Practicing what we teach: trauma-informed educational practice*	Trauma, Violence, & Abuse	2015	Foundational trauma-informed pedagogy text; flexible assessment rationale
IV-A	CAST	*Universal Design for Learning Guidelines 3.0*	CAST Technical Report	2024	UDL 3.0 update; asset-based approach; interdependence; barrier removal framework
IV-A	Hattie & Timperley	*The power of feedback*	Review of Educational Research	2007	Feedback theory; competence need satisfaction; Social Coherence via transparency
IV-A	Pan et al.	*AI chatbots as reading companions: a SDT perspective*	BJET	2025	SDT applied to AI reading support; autonomy, competence, relatedness in AI interaction
IV-B	Anderson & Heyne	*Flourishing through Leisure Model*	Therapeutic Recreation Journal	2012	TR theoretical foundation; five well-being dimensions aligned with Keyes SWB
IV-B	Rashid et al.	*PARO robot intervention: systematic review and meta-analysis*	Int. J. Nursing Studies	2023	Social robot evidence; SMD effects on depression/agitation; SDT relatedness
IV-B	Kim & Kim	*AI conversational agents among low-income older adults*	SAGE Open	2024	Qualitative evidence; reduced loneliness; emotional warmth; social integration
IV-B	Cheng et al.	*Gamified interventions in mental health: meta-analytic review*	Computers in Human Behavior	2023	Gamification evidence base; Hedges' g=0.38; engagement-efficacy gap
IV-B	PLOS Digital Health	*Immersive VR for older adult well-being: meta-analysis*	PLOS Digital Health	2026	IVR RCT synthesis: SMD=-0.83 for depression; 100% stress improvement rate
IV-B	MDPI	*Gamification: Engagement-Efficacy-Ethics trilemma*	Information (MDPI)	2025	Ethics critique of gamification; manipulation risk; Social Acceptance implications
MOD V	Ragnedda	*The Third Level of the Digital Divide (book extract)*	Routledge / Digital Journalism	2017	Third-level divide; who benefits from being online; digital social capital
MOD V	Zhao et al.	*Digital social capital and student development in higher education*	Nature Communications	2022	Social capital formation in digital contexts; bonding vs. bridging capital
MOD V	Dastyari & Jose	*Digital inclusion and achieving SDG 4*	Education in Law Journal (SAGE)	2024	SDG 4 + digital inclusion; inclusive and equitable quality education evidence

Module	Author(s)	Title (abbreviated)	Journal	Year	Relevance to Framework
MOD V	OECD	*AI, equity and inclusion in education*	OECD Report	2024	Algorithmic bias taxonomy; capability-constrained AI; policy recommendations
MOD V	Chang, Choi, Bazarova & Lockenhoeff	*Age differences in online social networking — SST extension*	J. Broadcasting & Electronic Media	2015	SST applied to SNS; smaller but more meaningful networks in older adults
MOD V	Keyes	*Mental illness and mental health: The two continua model*	J. Health & Social Behavior	2005	Flourishing/languishing distinction; lifespan application; capacity rebuilding outcome
MOD V	Navarro & Tudge	*AI and neo-ecological theory: Extended implications*	Current Psychology	2025	Forward-looking: AI as developmental agent in future community well-being architecture

D.2 SSCI Search Keywords by Variable 文獻搜尋關鍵字

Variable	Primary Search Keywords	Target Journals	Timeframe
Crisis Disruption (IV)	*"Educational disruption"; "Emergency remote teaching"; "Technostress"; "Systemic shock"; "Pandemic learning loss"; "Crisis pedagogy"*	Computers in Human Behavior; BJET; Computers & Education	2020–2026
Digital Mobilization (Mediator)	*"Digital agency"; "Social media mobilization"; "Community informatics"; "Technological agency"; "Learning analytics empowerment"; "AI-driven resource allocation"*	J. Computer Assisted Learning; Internet and Higher Education; IRRODL	2020–2026
Sustainable Adaptation [Track A]	*"Resilient pedagogy"; "Trauma-informed teaching"; "Universal Design for Learning"; "Pedagogy of care"; "Flexible assessment"; "Micro-learning"*	BJET; ETR&D; Teaching and Teacher Education	2022–2026
Sustainable Adaptation [Track B]	*"Digital therapeutic recreation"; "Serious games well-being"; "Gamified social space"; "VR leisure"; "Digital flow"; "Leisure-based coping"*	Leisure Sciences; Applied Research in Quality of Life; Therapeutic Recreation Journal	2020–2026
Capacity Rebuilding (Outcome)	*"Digital resilience"; "Social capital restoration"; "Social well-being digital"; "Community empowerment technology"; "Digital literacy social capital"; "Keyes social well-being"*	Internet and Higher Education; Social Indicators Research; J. Happiness Studies	2020–2026

Closing Remarks 結語

Research Gap This Course Addresses: No published study has simultaneously tested the complete mediation pathway of "microsystem → social support → social well-being" within a graduate-level digital learning environment. This course (VE5915701) and its associated Teaching Practice Research (TPR) provide an opportunity to generate such empirical evidence.

The Modular Flexibility Principle: The dual-track architecture (Track A: Resilient Pedagogy; Track B: Digital TR) is designed so that learners from different disciplinary backgrounds can find a primary intellectual home within the course without sacrificing a unified theoretical framework. Both tracks achieve the same Course Learning Outcomes (CLOs) through different empirical domains.

From Victim to Architect — The Central Metaphor: The core aspiration of the course is to transform learners' relationship with digital disruption: from passive recipients to active agents, from constrained capabilities to expanded functioning, and from social isolation to social actualization. This is not merely a pedagogical objective but also a political one—aligned with the emancipatory vision of Sen's Capability Approach and the Sustainable Development Goals (SDGs) commitment to "leave no one behind."

Version & Revision: This course design document is intended as a living framework. The SSCI reading list should be updated annually, and the weighting of Track A and Track B adjusted in response to emerging research evidence and community needs. The Teaching Practice Research (TPR) protocol should be rerun with each cohort to build a longitudinal evidence base.